Delta-SVD

Efficient Compression for Personalized Text-to-Image Models

POST-HOC COMPRESSION / NO RETRAINING / NO SUBJECT IMAGES

151 MiB

per-subject adapter at the SD-v1.5 knee

98.3%

of full-model mean DINO · 30 subjects

0 training steps

compress a finished checkpoint and its base

01 Compress the change

Personalization creates many large checkpoints. Store a shared base once, then a compact weight delta for each subject.

1 SUBTRACT

$$\Delta W = W_{\text{ft}} - W_0$$

↓

2 FACTORIZE

$$\Delta W = U \Sigma V^{\top}$$

↓

3 KEEP TOP COMPONENTS

$$\widehat{\Delta W} = U_r \Sigma_r V_r^{\top}$$

↓

4 MERGE AT LOAD TIME

$$\widehat{W} = W_0 + \widehat{\Delta W}$$

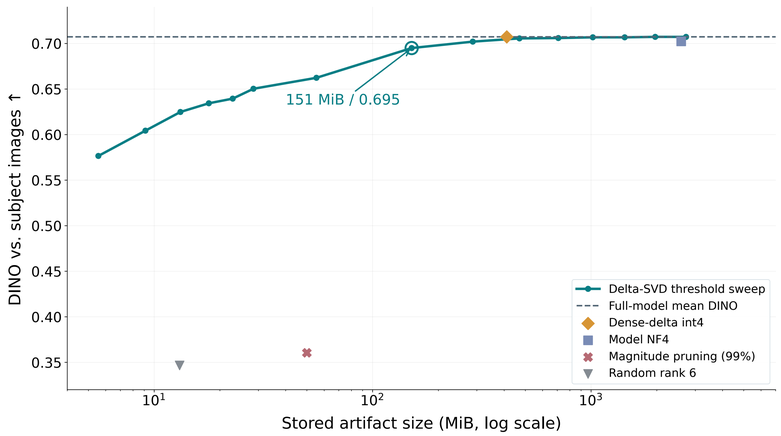
Choose the rank

Fixed rank, a per-layer spectral threshold, or a global byte budget. A rank-r factorization stores r(m + n) values per matrix.

Dense weights are reconstructed once at load time. Inference uses the original network graph; storage savings do not imply lower runtime VRAM.

02 Storage vs. subject fidelity

SD-v1.5 · DreamBooth · 30 subjects × 5 images per variant



Method / setting	MiB	DINO ↑
Full checkpoint	4,993.9	0.7073
Delta-SVD · t = 0.5	708.3	0.7060
Delta-SVD · t = 0.2	150.9	0.6950
Dense-delta int4	411.4	0.7073

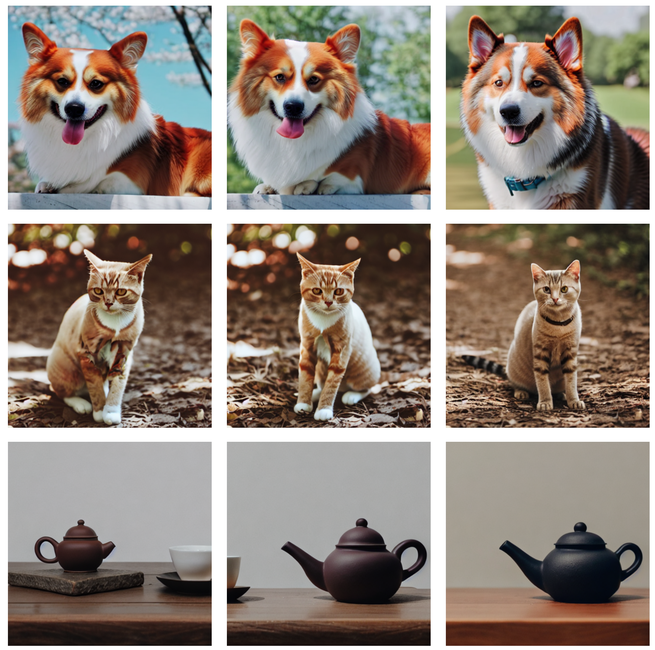
Near-full fidelity at 708 MiB

Mean paired DINO change: -0.0013; 95% subject-bootstrap CI [-0.00262, -0.00002] (n = 30).

MiB = 2^{20} bytes. Delta artifacts require the shared base; the full checkpoint includes pipeline components. Sizes are measured artifacts, not matched-component ratios.

03 What is retained?

Matched subject, prompt and image index in each row



Selected examples: aggressive truncation can change identity and shape. The 151 MiB setting better preserves the full-model appearance.

Beyond SD-v1.5

SD3.5-medium: 12 fully fine-tuned subjects; DINO 0.7427 → 0.7345 at 31.2 MB. Architecture and recipe vary; this is not a scaling law.

READING THE EVIDENCE

t denotes the implemented cumulative-singular-value threshold (Σσ), not squared energy (Σσ²). Budget allocation optimizes a spectral proxy; heterogeneous integer budgets do not have a general greedy-optimality guarantee.

Scope: per-subject storage; full fine-tuning results shown. Merged-LoRA SDXL / FLUX results are separate evidence. Sources: project result tables and generated images; full provenance and rebuild script accompany this poster.